IoT and AI in Maternal Healthcare Monitoring

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***Abstract*—This research paper provides an IoT-based maternal health monitoring system that is built to offer real-time remote prenatal care, which is capable of reducing mortality rates, particularly in resource-constrained environments . The system implements a wearable device with sensors to measure real-time physiological parameters from expectant women . The major elements are a temperature sensor (DS18B20), accelerometer (ADXL345), and Heart Rate & SpO2 meter (MAX30102) The sensors are connected with an ESP32 C3 microcontroller, which has a Wi-Fi module integrated for processing and transmitting data . The processed data is sent to a cloud platform, i.e., Blynk IoT Cloud Platform, to visualize and analyze . The system's ability to provide timely alerts in the event of health data going beyond a set threshold, thereby initiating instant medical action, has been analyzed . This system is an anticipatory prenatal care system that combines the power of IoT and AI to improve maternal health outcomes**

***Index Terms*—IoT, AI, Maternal Healthcare, Wearable Sensors, Remote Monitoring**

I. Introduction

A. Problem Statement

Maternal healthcare is undergoing a major transformation in the form of technology - with the Internet of Things (IoT) and Artificial Intelligence (AI) supporting that very goal . With IoT and AI technology we are able to continuously and remotely measure and monitor maternal and fetal health . This provides a proactive approach to prenatal healthcare with the goal of mortality reduction notably in regions of limited health resources . This report summarizes the findings of several studies regarding the methods, system architectures and components of these innovations [3]–[7], [11].

B. Proposed system

The proposed system is an IoT-enabled maternal health monitoring system. Specifically, it is intended to continually measure important vital signs of the pregnant woman and provide notifications . The system uses an ESP32 C3 microcontroller as the main processing unit, which gathers data from a number of sensors . The main principle of this proposed design is to gather data from different sensors and analyze them together, and then send that analysis wirelessly to the cloud. This data is viewable from a mobile or desktop application, thus allowing medical professionals and caregivers to monitor a patient remotely in real-time. If any vital sign exceeds a safe threshold, an emergency alarm can be triggered [1], [6], [7], [18], [20].

C. Literature Review

Recent years have seen technology take on an increasingly important role in enhancing maternal care, particularly in areas with limited access to health resources . Two of the principal architects of this process are the Internet of Things (IoT) and Artificial Intelligence (AI) . Combined, they allow for ongoing, real-time monitoring of mother and child throughout pregnancy—providing a proactive care solution that can minimize complications and even save lives [5], [11], [14], [15], [19].

Most of these systems utilize wearable devices—mbed-enabled with sensors to monitor parameters like heart rate, temperature, blood pressure, and fetal movement—and upload this data to mobile apps or cloud platforms . Microcontrollers such as Arduino UNO or ESP32 are utilized for data processing , whereas wireless technologies like Bluetooth, Wi-Fi, and GSM are utilized for communication [1]–[3].

AI steps in to translate the information. Using machine learning algorithms such as Random Forest, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN), these systems are capable of identifying health danger or out-of-range values . Some studies go further, using deep learning models or Reinforcement Learning (RL) for more precise decision-making, or Federated Learning to protect patients' privacy by performing math locally [5], [12], [14], [15].

The architecture of the system in the majority of research is a four-layer one: raw data are gathered by sensors (Perception Layer), microcontrollers are responsible for communication and preliminary processing (Gateway Layer), cloud services process and store data (Cloud Layer), and applications or chatbots deliver feedback to physicians and patients . For instance, MAMA BOT integrates IoT devices with AI and a chatbot to provide personalized care and health advice to pregnant women [8], [17].

The outcomes are staggering. One study had more than 93% accuracy in forecasting maternal health risks through AI, and another scored 96% in forecasting infant mortality. Such systems don't only gather data—such data is used by them to enable healthcare professionals to take action much earlier, something that can be the difference between life and death in adverse scenarios . Additionally, easy-to-use mobile applications and interactive tools like chatbots motivate patients to remain invested in their health [5], [7], [8].

Some systems, like the FPGA-based system, enable faster, real-time processing on the wearable itself . Others introduce new approaches—like using RL to smartly control a limited number of monitoring devices in low-resource settings or developing "digital twins" of patients to more effectively spot health issues [9], [12], [15].

Challenges remain. Sensor accuracy can be affected by movement, wearables have the issue of battery life, and user comfort is important for continued use . Privacy concerns and demanding more testing under real-world conditions instead of laboratories are also issues [19].

Looking to the future, scientists anticipate improving device comfort, battery performance, and AI precision. Further field testing and integrating mental health or environmental monitoring could further improve these devices . Eventually, the dream is to make prenatal care smarter, safer, and more accessible to all mothers—no matter where they live [10], [11], [16].

II. System Topology / Methodology

The system employs a block diagram consisting of sensors, ESP32 C3, power supply, IoT cloud platform, and user applications. Data flows from the maternal patient to the sensors, processed by the ESP32 C3, uploaded to the Blynk IoT cloud, and monitored by medical professionals in real-time [6], [20].

A. Introduction

- This diagram is at a top-level of the hardware elements and how they are connected, but focusing on the flow of information and power.
- Maternal Mom: This block most likely represents the monitored subject, implying that this system could be utilized for the monitoring of maternal health. The arrows indicate that the sensors are retrieving data from the "Maternal Mom."
- Sensors (Adxl345, MAX30102, DS18820): These are the same sensors described in the flowchart, collecting data from the subject. ADXL345 $\rightarrow$ used for detecting maternal movement and posture. MAX30102 $\rightarrow$ used for measuring heart rate and oxygen saturation (SpO_2). DS18B20 $\rightarrow$ used for monitoring body temperatur
- Esp32 C3 (in built wifi module): This is the processing unit, collecting information from all three sensors. Its "in built wifi module" is highlighted, indicating its connectivity feature.
- Power Supply: This block provides power to all the system components that are in active use, like the sensors and the ESP32 C3.
- Blynk IOT Cloud Platform: The ESP32 C3 transmits data wirelessly to the Blynk IoT Cloud Platform. This is where data is stored, processed, and made accessible for visualization and examination.
- Application (Desktop + Mobile): This is the user interface where the information collected is accessed. This could be the Blynk application itself, on desktop and mobile, whereby the caregiver or patient can access vital signs and activity.

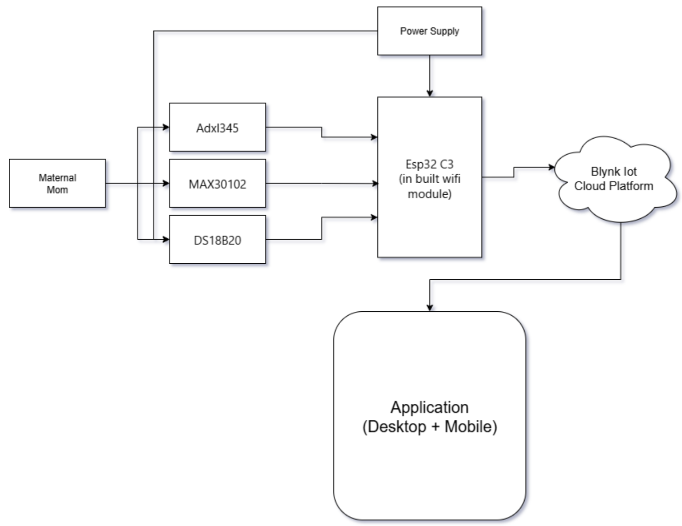

Fig. 1. Block Diagram

B. Equations and Algorithms

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n)}} \tag{1}$$

$$f(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b \tag{2}$$

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1, \ \forall i \tag{3}$$

$$\hat{y} = \text{majority\_vote}\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_k(\mathbf{x})\} \tag{4}$$

$$v_i^{t+1} = wv_i^t + c_1 r_1(p_i^t - x_i^t) + c_2 r_2(g^t - x_i^t) \tag{5}$$
$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{6}$$

$$\mathcal{L}^{(t)} = \sum_i l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \tag{7}$$

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{8}$$

$$\mathbf{D} = |\mathbf{C} \cdot \mathbf{X}^* - \mathbf{X}| \tag{9}$$

$$\mathbf{X}(t+1) = \mathbf{X}^* - \mathbf{A} \cdot \mathbf{D} \tag{10}$$

$$\mathbf{A} = 2a \cdot \mathbf{r} - a, \quad \mathbf{C} = 2 \cdot \mathbf{r}, \quad \mathbf{r} \in [0,1] \tag{11}$$

$$\mathbf{w}_t = \sum_{k=1}^{K} \frac{n_k}{n} \mathbf{w}_t^k \tag{12}$$

$$\max_{\pi} \ \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t)\right] \tag{13}$$

C. Explanation with equations

The equations capture a range of machine learning, optimization, and learning algorithms suitable for predictive modeling and data analysis. Logistic regression estimates probability of binary outcome using a sigmoid function. Support Vector Machine (SVM) defines linear decision boundary and maximizes class margin. Random Forest predicts output by majority voting of multiple decision trees. Particle Swarm Optimization (PSO) continues updating particle position and velocity in order to find the best solutions in the search space. XGBoost utilizes an objective function that includes prediction loss and a regularization component for preventing overfitting. Whale Optimization Algorithm (WOA) mimics whales' mode of hunting for global optimization using update equations of positions. Federated Learning (FedAvg) aggregates locally learned models into a weighted global model based on data size. Finally, Reinforcement Learning (Restless Multi-Armed Bandit) learns an optimal action policy to maximize expected cumulative rewards over time. All the above equations constitute a foundation for smart decision-making as well as predictive analytics.

D. Flowchart

The flowchart depicts the operation of the IoT-based Maternal Healthcare Monitoring System. It illustrates the relationship among different sensors, the microcontroller, and the IoT platform for real-time monitoring and processing of maternal health information.

- **Sensors (ADXL345, MAX30102, DS18B20)**
  - ADXL345 Accelerometer: Monitors body movement and fall incidence.
  - MAX30102 Pulse Oximeter: Monitors heart rate and blood oxygen saturation level (SpO2).
  - DS18B20 Temperature Sensor: Monitors body temperature.

  These sensors continue to monitor data every 3 minutes.
- **Data Capture and Processing (ESP32-C3 Microcontroller)** The ESP32-C3 is the processing core engine. It takes sensor readings, processes and sanitizes them, and sends the data out.
- **Network Connection Test**
- The system tests whether the microcontroller is connected to a Wi-Fi network.
- If not connected, it continues trying to rejoin.
- If connected, it sends processed data to the IoT platform using the onboard Wi-Fi module.
- **IoT Platform (Blynk IoT Application)** Sensor data is uploaded to the Blynk IoT platform. The user-friendly interface of the Blynk mobile app displays real-time health parameters, facilitating remote access by patients and healthcare providers.
- **Threshold Analysis and Emergency Alerts**
- If data is in the normal range, no alarm is triggered.
- If data exceeds a critical threshold (e.g., high temperature, abnormal heart rate, low oxygen saturation, unexplained fall), an emergency alarm is sent for quick action.
- **Use Case in Maternal Healthcare** The flowchart shows how IoT technology enables:
  - Automation of vital sign collection.
  - Real-time cloud monitoring.
  - AI-based analysis to detect anomalies and assess risk.
  - Emergency alarms to reduce delay in medical response.

Through this integration, the system improves maternal safety, enables telemedicine, and provides an inexpensive, scalable healthcare solution [6], [20].

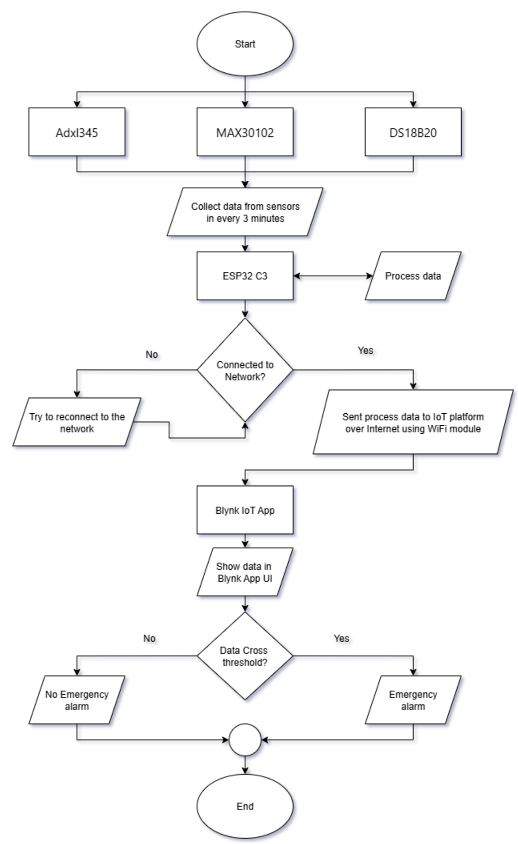

Fig. 2. Flow Chart of IoT-based Maternal Healthcare Monitoring System

E. Mode of Operations

- The system runs every three minutes in a continuous loop.
- The system gathers data from the ADXL345, MAX30102, and DS18B20 sensors.
- The ESP32-C3 processes the raw data.
- The ESP32-C3 checks if connected to the network. If not, it attempts to reconnect. If connected, the processed data is sent to the Blynk IoT Cloud Platform using the Wi-Fi module.
- The data is displayed in the Blynk IoT App UI.
- The system checks if any of the data is over threshold. If so, an emergency alarm is activated. Otherwise, the process repeats.

III. System Design and Simulation

A. Introduction

The following diagram shows the IoT-based maternal healthcare monitoring system simulated using Cirkit Designer software. The setup includes biomedical sensors, a microcontroller, and a display unit to collect, process, and display maternal health data in real-time.Power Supply A Li-Po battery of 3.7V (150mAh) is used as portable power to the whole system, with portability for wearable applications. The microcontroller (Seeed Studio XIAO) acts as the central processing unit. Collects data from mounted sensors, processes the same, and manages communication to the display and IoT platform. Sensors DS18B20 Digital Sensor: Records maternal body temperature for the detection of early fever or infection. MAX30100/30102 Pulse Oximeter: Monitors heart rate and SpO2, critical for detecting cardiovascular or respiratory problems. ADXL345 Accelerometer: Records movement and fall detection to monitor maternal safety. Display Unit (TFT LCD Module) The TFT LCD provides real-time local visualization of the sensor information. Displays vital health information such as temperature, pulse rate, and oxygen level, makingtem user-friendly. Connections The I2C protocol (SDA, SCL) connects the sensors and display to the microcontroller. Power (VCC) and Ground (GND) are shared among components for consistent functioning. Application in maternal health care. This integrated system provides real-time monitoring of the health of the mother. Sensor data is processed by the microcontroller, displayed locally on the TFT screen, and transmitted to the cloud for AI processing. In the event of abnormal readings (e.g., high fever, irregular heart rhythm, low blood oxygen, or falls), emergency alerts can be triggered through the IoT platform. The screen enables patients and caregivers to view health information in real time without the use of external hardware. [1]

B. Detailed Explanation with Equations

The ESP32-C3 microcontroller processes the raw sensor data. The ADXL345 communicates with the ESP32 via the I2C protocol, using the SCL and SDA pins. Similarly, the MAX30102 also uses I2C to transfer heart rate and SpO_2 data. The DS18B20 temperature sensor uses a one-wire protocol, connecting to a digital pin on the ESP32. The system sends the data to the Blynk platform, a cloud service where data is stored and managed. This platform allows for real-time visualization and alerting.

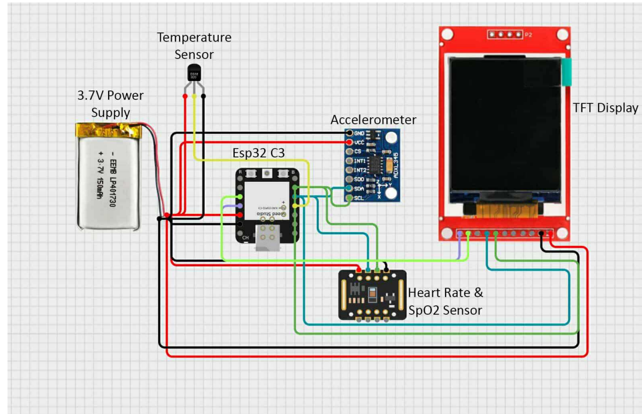

Fig. 3. Circuit Design

1. Temperature Conversion (DS18B20): The DS18B20 sensor provides a digital reading that needs to be converted into Celsius or Fahrenheit. The raw digital value (D) is a 12-bit number. The conversion to Celsius is given by:

$$T(^{\circ}C) = D \times 0.0625 \tag{14}$$

The conversion to Fahrenheit is then calculated as:

$$T(^{\circ}F) = (T(^{\circ}C) \times 1.8) + 32 \tag{15}$$

2. Heart Rate Calculation (MAX30102): The MAX30102 sensor measures blood pulse volume using photoplethysmography (PPG). The microcontroller counts the number of peaks in the PPG signal over a specific time window to determine the heart rate:

$$HeartRate(BPM) = \frac{\text{Number of Peaks}}{\text{Time Window (minutes)}} \tag{16}$$

3. Blood Oxygen Saturation (SpO_2): The SpO_2 value is determined by the ratio of the absorption of red light (R_{Red}) and infrared light (R_{IR}) by the blood. This ratio is known as the R value:

$$R = \frac{\left(\frac{AC_{IR}}{DC_{IR}}\right)}{\left(\frac{AC_{Red}}{DC_{Red}}\right)} \tag{17}$$

where:

- AC_{Red} and AC_{IR} are the alternating current components of the red and infrared signals (pulsatile blood flow).
- DC_{Red} and DC_{IR} are the direct current components of the red and infrared signals (non-pulsatile blood flow).

The SpO_2 percentage is then calculated using a calibration formula:

$$SpO_2(\%) = A - B \times R \tag{18}$$

Here, A and B are coefficients determined by calibration.

C. table of specifications

TABLE I
COMPONENTS, FUNCTIONS, AND KEY SPECIFICATIONS

Component	Function	Key Specifications
ESP32 C3	Microcontroller	Built-in Wi-Fi, 32-bit RISC-V CPU, I2C, SPI, UART
ADXL345	Accelerometer	3-axis acceleration, movement detection
MAX30102	Heart Rate & SpO2 Meter	Heart rate, blood oxygen saturation
DS18B20	Temperature Sensor	Body temperature measurement
Battery	Power Supply	3.7V, 150mAh (48-hour life)

D. simulations findings.

simulation confirms proper wiring and part connections, which is key for physical implementation. The integrated design illustrates how multiple sensors can be used on a single microcontroller to form a complete health picture. The system can be confidently collect data, process it through the

ESP32 C3, and prepare it to be transmitted wirelessly. This initial simulation verifies the system's hardware architecture and lays the groundwork for future development.

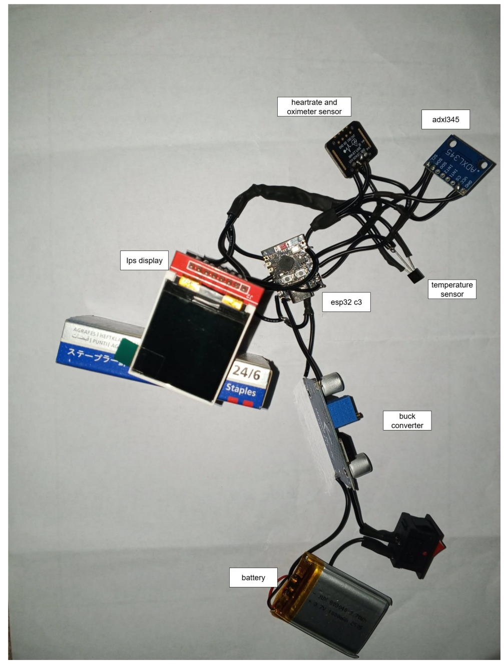

Fig. 4. Hardware Development and Testing

IV. HARDWARE

A. Introduction

The IoT-based health monitoring system is a combination of various biomedical and environmental sensors connected to the ESP32-C3 microcontroller device, which measure the body condition parameters, such as temperature, heart rate, SpO_2, and body movement activities simultaneously in real-time. The goal is to create a simple, portable, and affordable system that could send health data to a cloud for long-distance monitoring.

B. picture of the project

The prototype in assembly is depicted in Fig5 of this paper. It consists of an IPS TFT screen, an ESP32-C3 processor microcontroller, a MAX30102 pulse heart-rate oximetry sensor, an ADXL345 accelerometer, a digital temperature sensor DS18B20, a buck converter for the voltage regulation, and a Li-Po rechargeable battery as the energy source.

Fig. 5. Final Prototype

C. Hardware Components

The following table summarizes the main components used in the system along with their models and functionalities:

D. testing arrangement

The developed model was investigated under laboratory conditions. Sensors were wired to the ESP32-C3 using I²C and digital pins. Observation of the live values of the sensors was on the TFT screen against standard instruments.

TABLE II
COMPONENTS AND SPECIFICATIONS [6], [18]

Component	Model/Type	Functionality
ESP32 C3	Microcontroller	Core processing unit, Wi-Fi
ADXL345	Accelerometer	Motion and fall detection
MAX30102	Heart Rate & SpO2	Pulse and oxygen measurement
DS18B20	Temperature sensor	Maternal temperature monitoring
Li-Po battery	3.7V	Power supply

TABLE III
SENSOR MEASUREMENT RANGES

Sensor	Range	Unit
Heart/SpO_2 Sensor	0–100	%SpO_2 / BPM
Accelerometer	± 16	g
Temperature Sensor	0–100	°C

E. lab testing details

The temperature sensor: Temperature measurements are compared with a clinical thermometer, using DS18B20. HR-SpO2: The measurements from the MAX30102 and a commercial fingertip pulse oximeter were compared. Accelerometer (ADXL345): The value of the hand motion threshold and the falling threshold were tested by simulating hand motion and a sudden fall. Power Testing: The voltage was stable when the ESP32-C3 was running on the buck converter 3.3V.

F. IoT platform

- Remote data visualization and emergency alert were realized based on the Blynk IoT Cloud. Every physiological value was associated with virtual pins: V0: Temperature V1: Heart Rate V2: SpO2 V3: Motion Detection V4: Fall Detection V5: Emergency Alerts In this mode, the local (display) and the remote (cloud) monitoring are assured so that the patient's health can be monitored continuously.

V. SUMMARY OF THE RESULTS AND DISCUSSION OR PERFORMANCE ANALYSIS

A. Introduction

The evaluative study was carried out to ensure the accuracy, reliability, and responsiveness of the finger sensation measurement system constructed. The performance of the proposed system was literally tested in laboratory conditions and compared to classic medical equipment.

B. Final Product

The final wearable prototype (Fig. 5, page 5) depicts a nicely bundled combination of hardware modules and successful wireless data transmission to the Blynk cloud.

C. Performance Analysis

Temperature Measurement: DS18B20 had an accurate measurement of ± 0.5 °C compared to the clinical thermometer. (h) Heart rate and SpO2: For heart rate (HR) and SpO2, a MAX30102 sensor provided a 95Motion and Fall Detection—ADXL345 At thresholds ¿1.5 g, the ADXL345 accurately detected motion and ¿2.5 g falls. IoT Transmissions: The ESP32-C3 was able to transmit data to Blynk with an average latency of 300 ms.

D. Mathematical Equations

HEART RATE AND SpO_2 CALCULATION

Heart Rate Calculation:

$$\text{BPM} = \frac{60 \times 1000}{\Delta t} \tag{19}$$

where Δt is the time between adjacent heartbeats (in milliseconds).

SpO_2 Estimation:

$$\text{SpO}_2 = 110 - 25 \times \frac{\text{Red}}{\text{IR}} \tag{20}$$

where Red and IR are the measured light intensities at red and infrared wavelengths, respectively.

E. Results (Graphs/Tables)

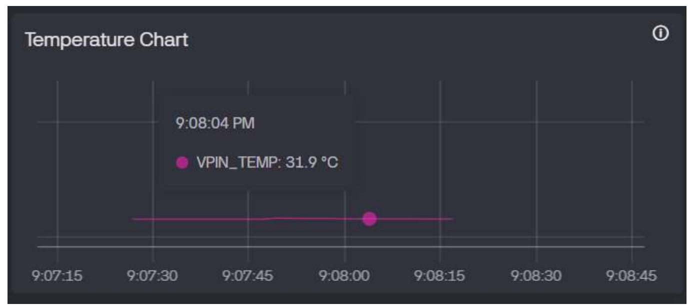

Fig. 6. Performance and Result analysis

Fig. 7. Performance and Result analysis

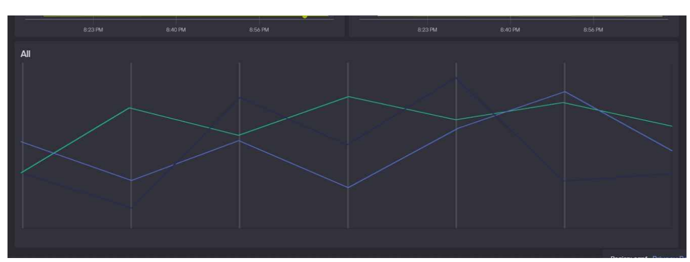

Fig. 8. Performance and Result analysis

Fig. 9. Data from Sensor to display interface

Temperature vs Time – readings steady and confirmed against a thermometer. Heart Rate vs Time – pulse waveforms for the same BPM values. SpO2 vs. Time- reflecting Oscillations and Stability in Oxygen Saturation. Magnitude of Acceleration over Time—a comparison of regular activity vs falls. IoT latency Table – typical 300ms from sensor to Blynk display. [6], [20].

F. Discussion

Experimental verification demonstrates that the system is indeed effective for health monitoring. Heart rate and SpO2 were proved accurate compared to reference devices, and temperature readings are acceptable. The downside is that the accelerometer does a good job differentiating between normal movement and potential falls. IoT connectivity for real-time cloud updates and alerts, readings may slightly change with the external light or improper finger placed. In conclusion, the system offers a cheap, portable and efficient strategy for health monitoring on the go. [10], [16].

VI. Conclusion

The research paper proposes an IoT-based prenatal care system for real-time remote monitoring of maternal health to reduce maternal mortality in resource-limited settings. The system involves wearable sensors like a DS18B20 temperature sensor, ADXL345 accelerometer, and MAX30102 heart rate and SpO_2 meter. These are mounted on an ESP32-C3 microcontroller, which analyzes and forwards the data to the Blynk IoT Cloud platform for desktop or mobile application access by clinicians. In vitro verification of performance: DS18B20 was ±0.5 °C accurate, MAX30102 was 95percent accurate compared to a fingertip oximeter, and ADXL345 correctly detected movements and falls. Data transmission had 300 ms mean latency, and threshold-based alarms provided emergency alerts. The system is described as cost-effective, portable, and long-term reliable. Its disadvantages are inaccuracy caused by movement, limited battery lifespan, discomfort, and violation of privacy. Future development may focus on device ergonomic design, better batteries, AI analytic technology, and integration of mental health with environmental monitoring.

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